

L2 - Graphing Sine and Cosine Functions

MCR3U

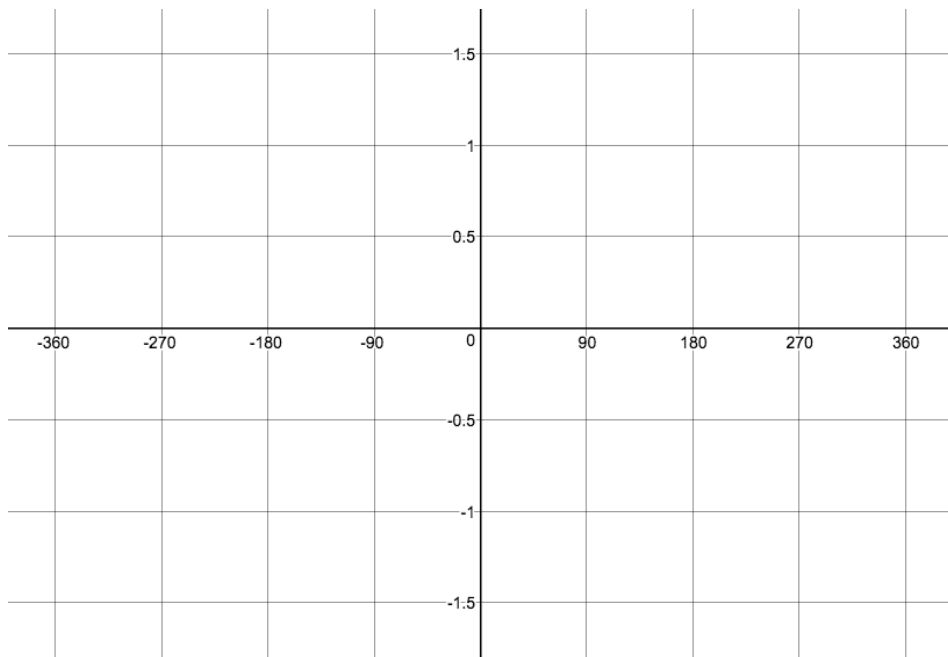
Section 1: Graphing Sine and Cosine

[DESMOS demonstration](#)

To graph sine and cosine, we will be using a Cartesian plane that has angles for x values.

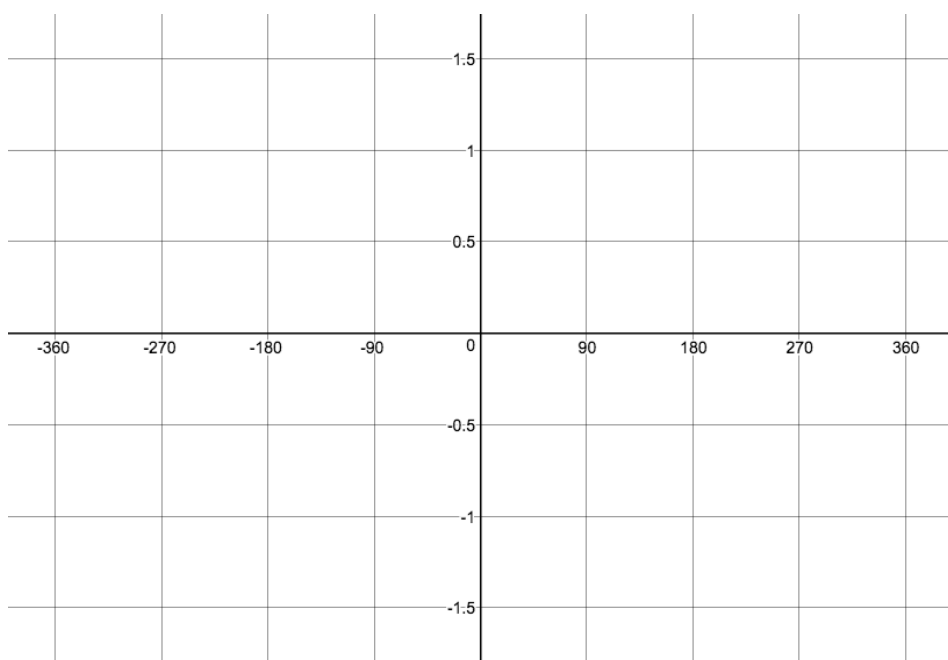
Example 1: Complete the following table of values for the function $f(x) = \sin(x)$. Use special triangles, the unit circle, or a calculator to find values for the function at 30° intervals. Use the table to graph the function.

x	$f(x)$
0	
30	
60	
90	
120	
150	
180	
210	
240	
270	
300	
330	
360	



Example 2: Complete the following table of values for the function $f(x) = \cos(x)$. Use special triangles, the unit circle, or a calculator to find values for the function at 30° intervals. Use the table to graph the function.

x	$f(x)$
0	
30	
60	
90	
120	
150	
180	
210	
240	
270	
300	
330	
360	



Section 2: Properties of Sine and Cosine Functions

Domain:

Range:

Period:

Amplitude:

Section 3: Transformations of the Sine and Cosine Functions

$$y = a \sin[k(x - d)] + c$$

[Desmos Demonstration](#)

a	k	d	c
Vertical stretch or compression by a factor of a . Vertical reflection if $a < 0$ $ a = \text{amplitude}$	Horizontal stretch or compression by a factor of $\frac{1}{k}$. Horizontal reflection if $k < 0$. $\frac{360}{ k } = \text{period}$	Phase shift $d > 0$; <i>shift right</i> $d < 0$; <i>shift left</i>	Vertical shift $c > 0$; <i>shift up</i> $c < 0$; <i>shift down</i>

Example 3: For the function $y = 3 \sin[2(\theta + 60^\circ)] - 1$, state the...

Amplitude:	Period:
Phase shift:	Vertical shift:
Max:	Min: